# From SSO to NGS in HLA population studies: the case of Mandenka (*Senegal*)

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# Genetic data & Typing methods

## Genetic data

**Mandenka** (*Niger-congo*)
Western Africa – Senegal

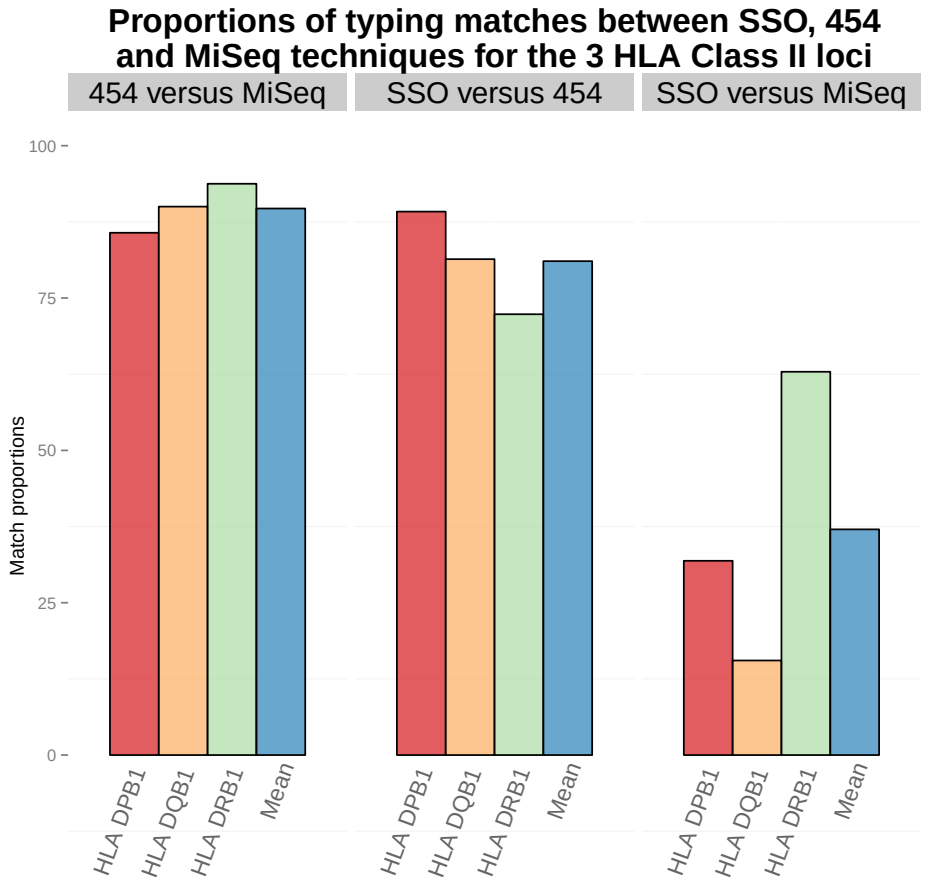
3 **Class II HLA** genes:
HLA-DPB1 ; -DRB1 ; -DQB1

## Typings

1. PCR-SSO (Geneva 1992)
2. NGS:
   - **Exon 2**: 454 pyrosequencing (Geneva 2012)
   - **Full gene**: MiSeq (Stanford and Geneva, 2015-2016)

**Figure :** Sampling location in the Kedougou city neighborhood, eastern Senegal

# Comparisons of techniques

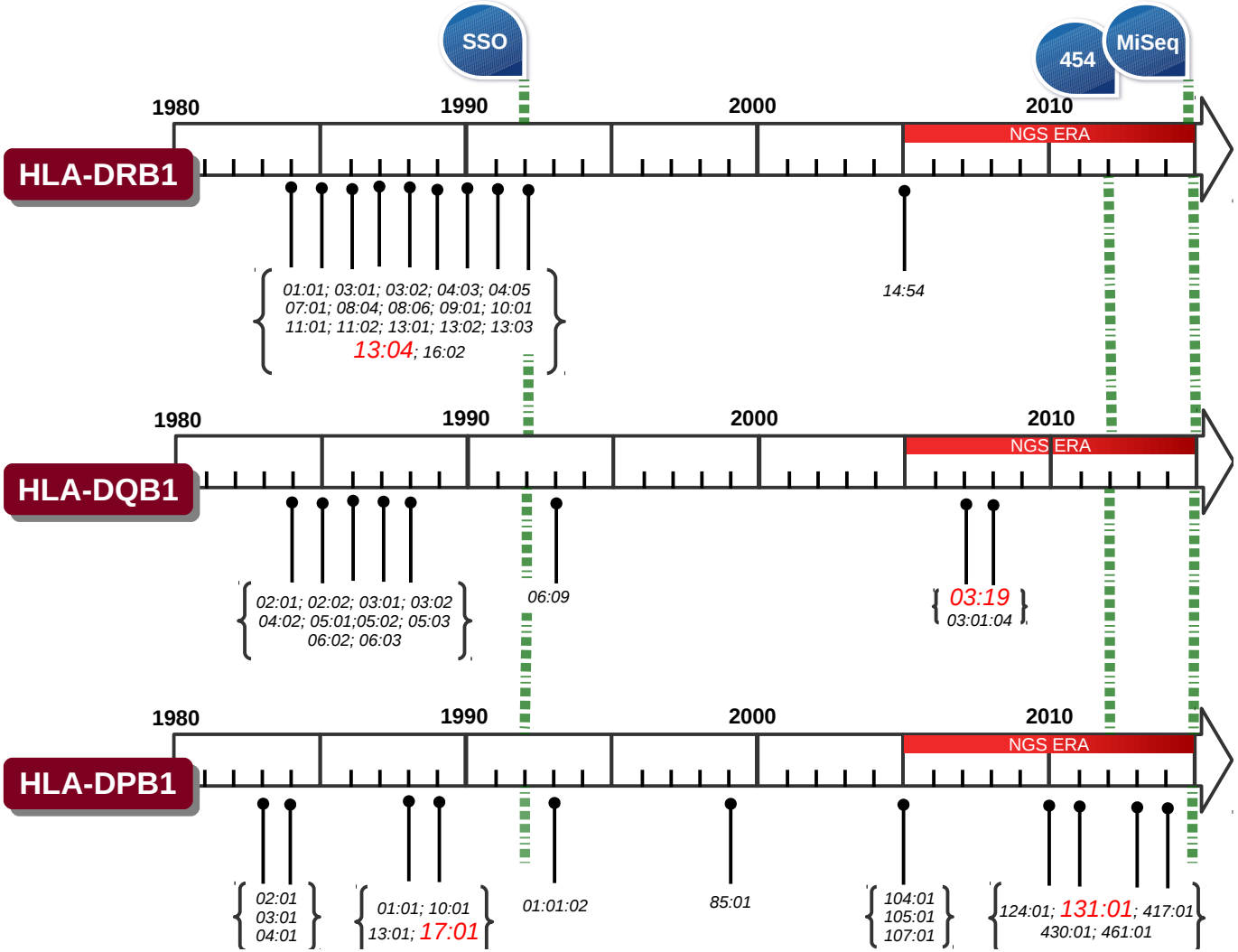



Good match between NGS:
**DNA sequences**

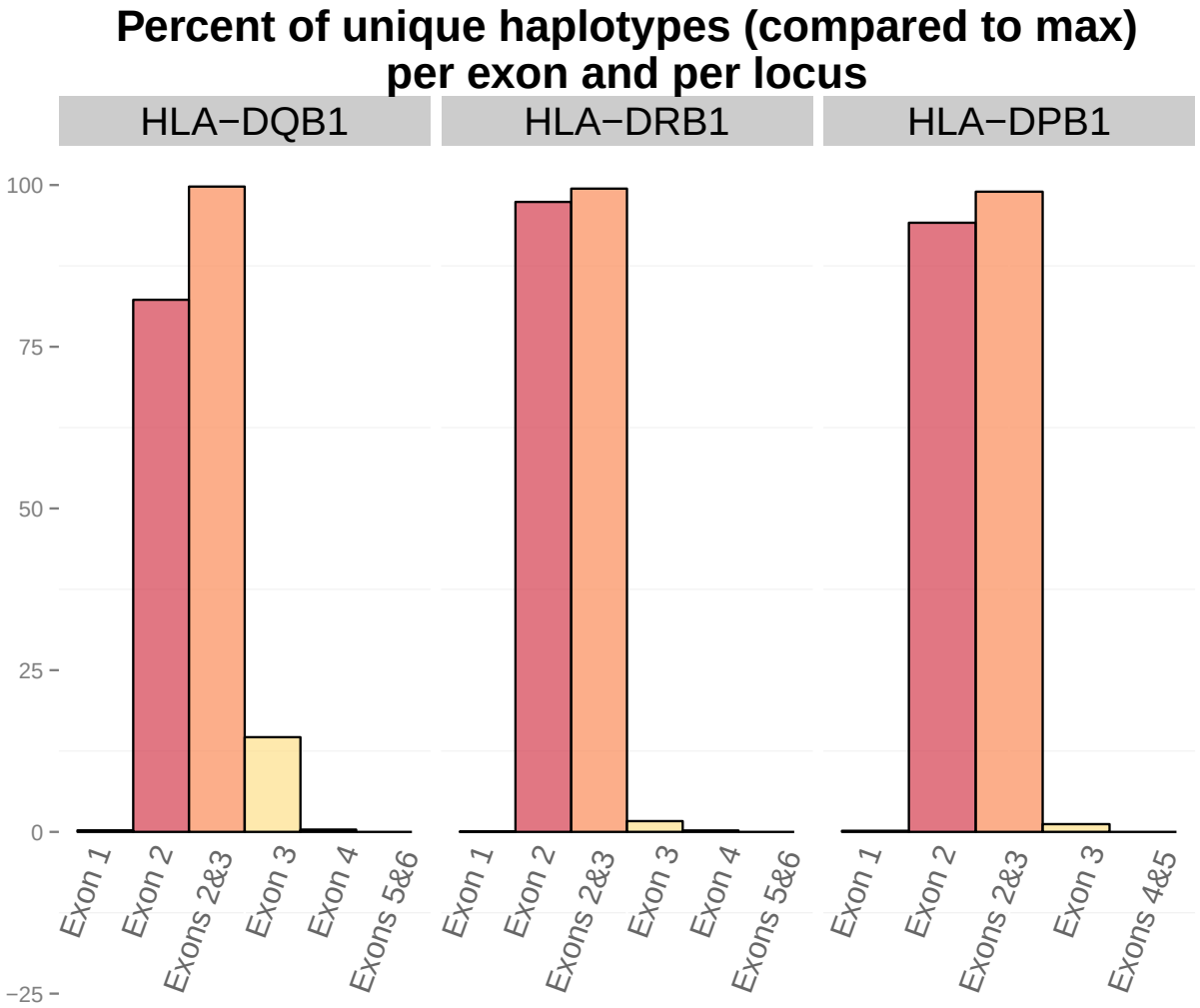
Correct match between 454 & SSO:
**Exon 2**

Discrepancies between SSO & MiSeq:
**Locus-dependant**

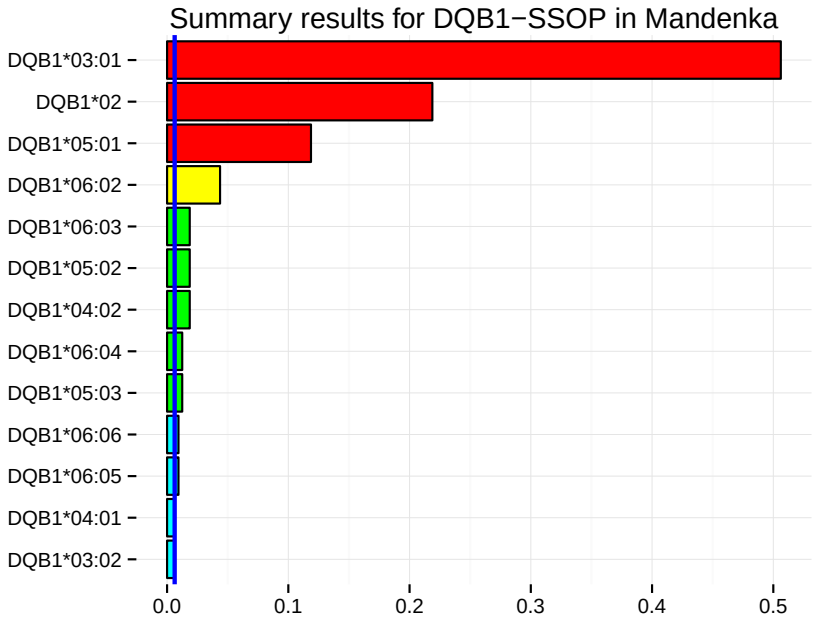
# Years of allele reports



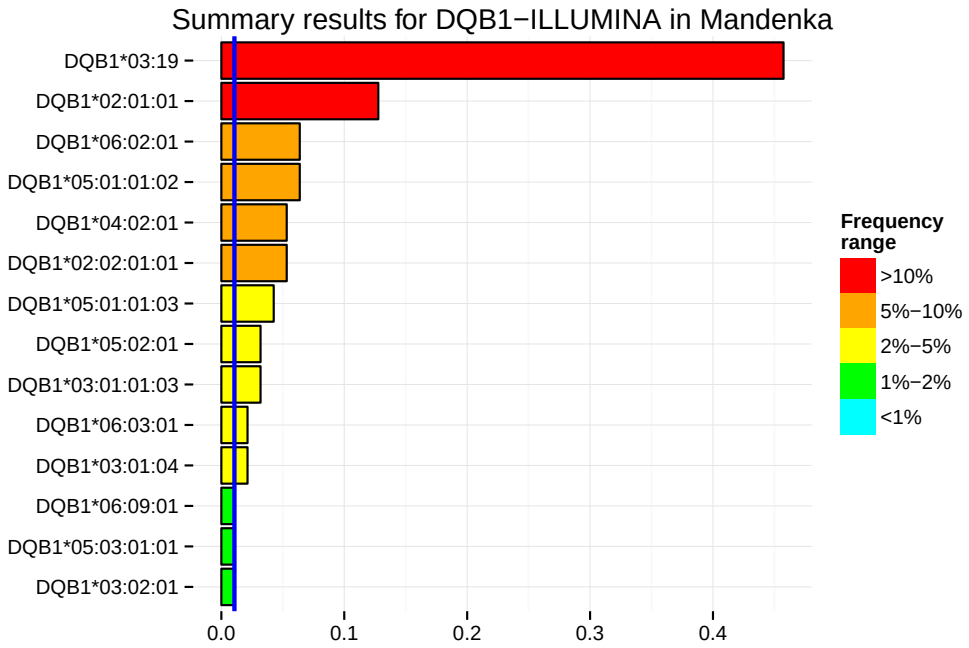
# Resolutive power of Class II HLA exons



# HLA-DQB1: *Frequencies in Mandenka*

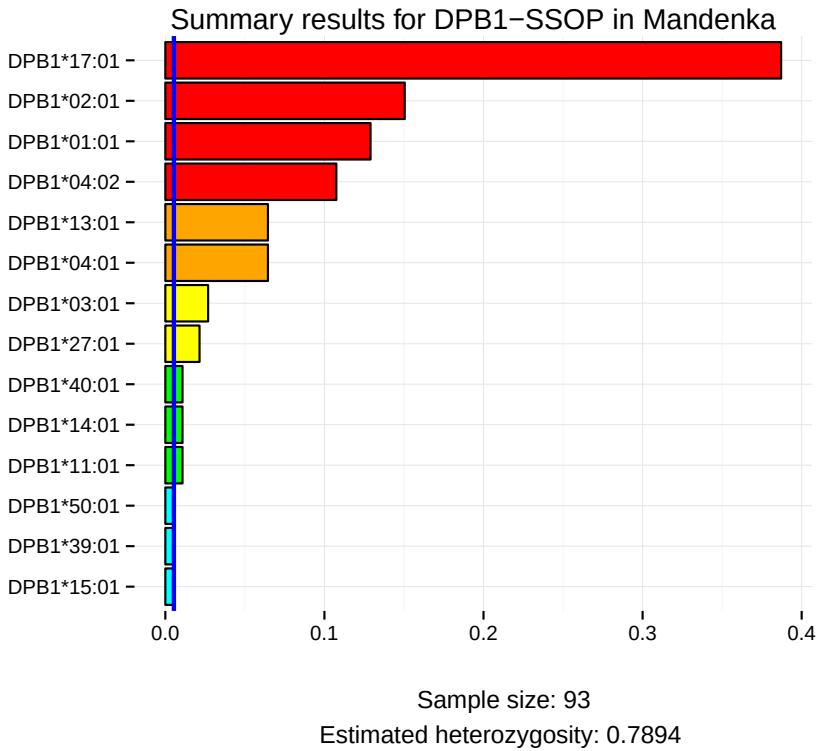


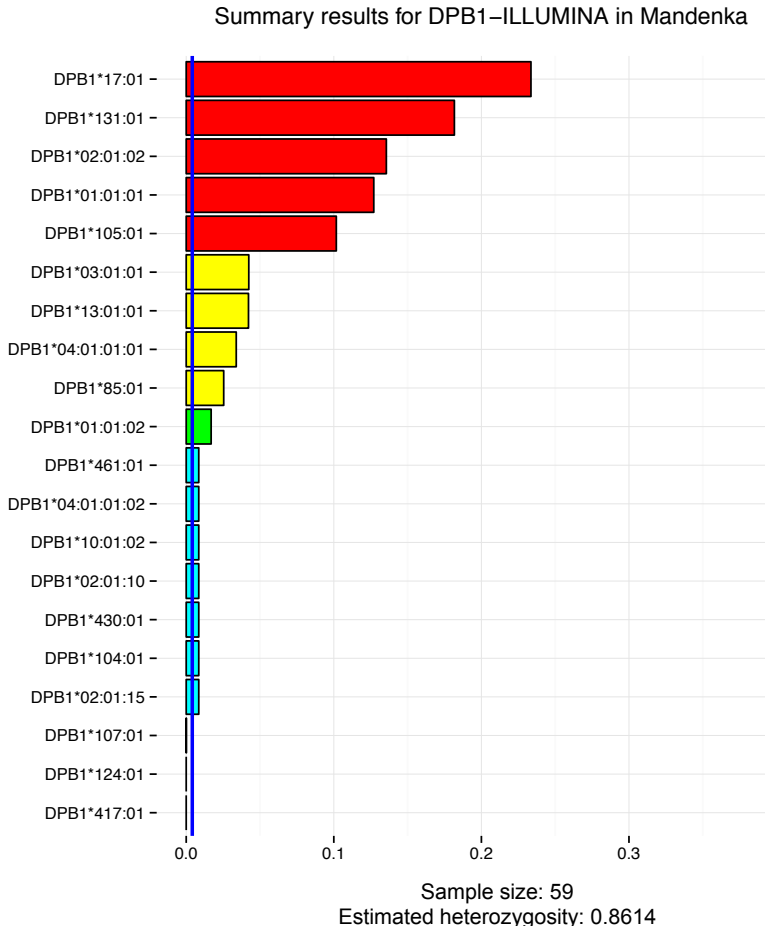

Sample size: 80
Estimated heterozygosity: 0.6783



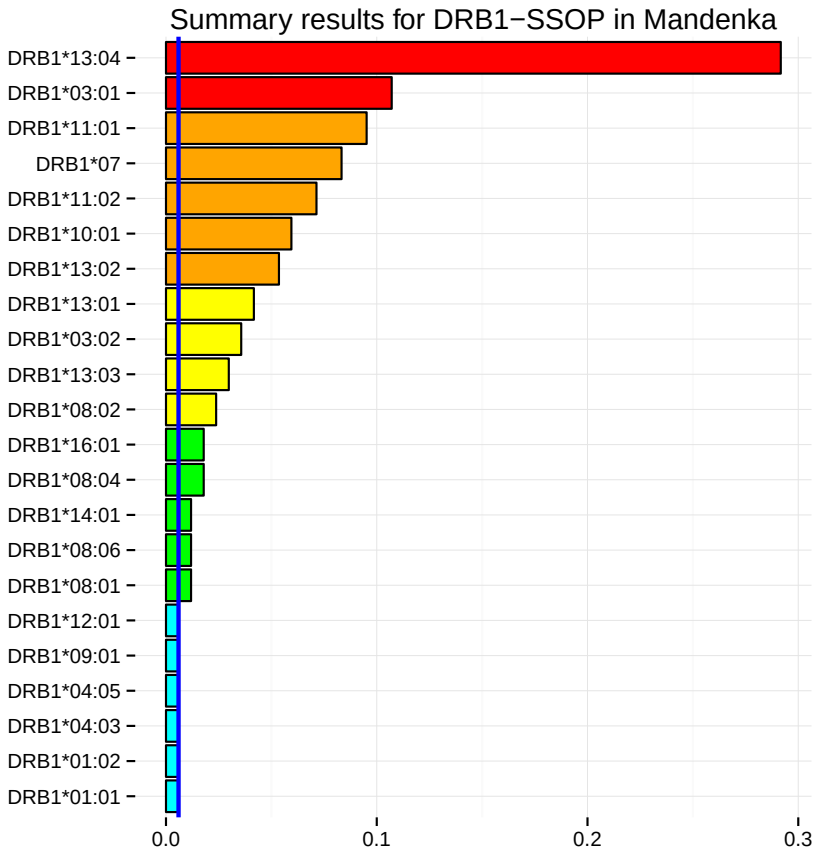

Sample size: 47
Estimated heterozygosity: 0.7556

# HLA-DPB1: *Frequencies in Mandenka*

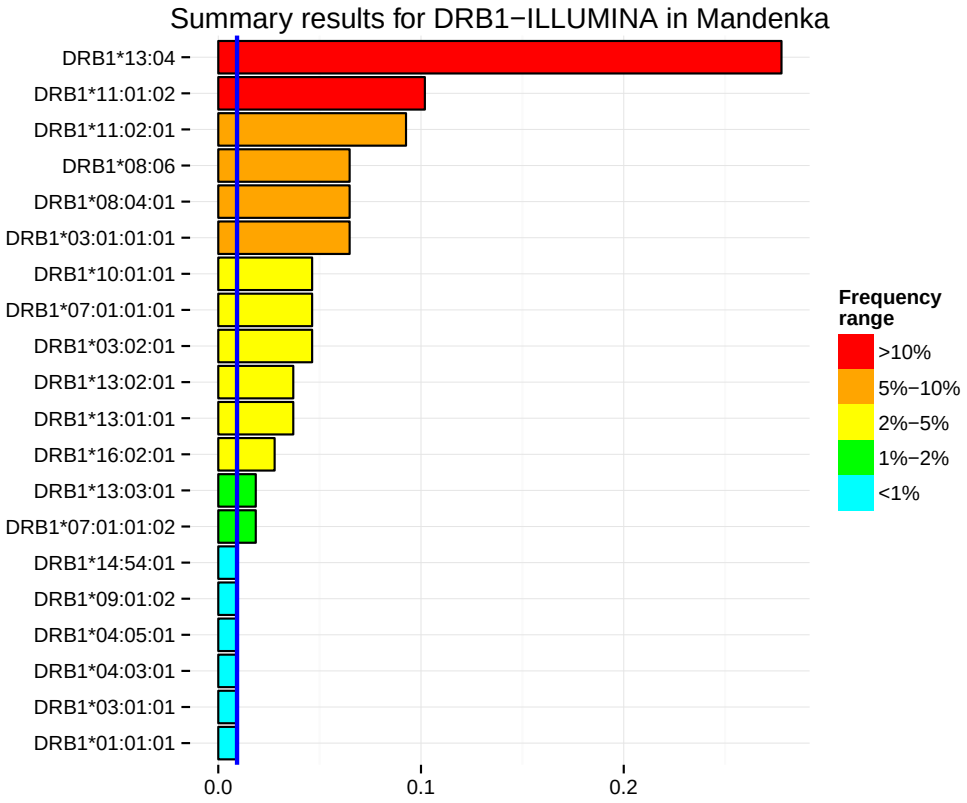


# HLA-DRB1: *Frequencies in Mandenka*




Sample size: 84
Estimated heterozygosity: 0.8702




Sample size: 54
Estimated heterozygosity: 0.8801

# Haplotypic and global linkage disequilibrium

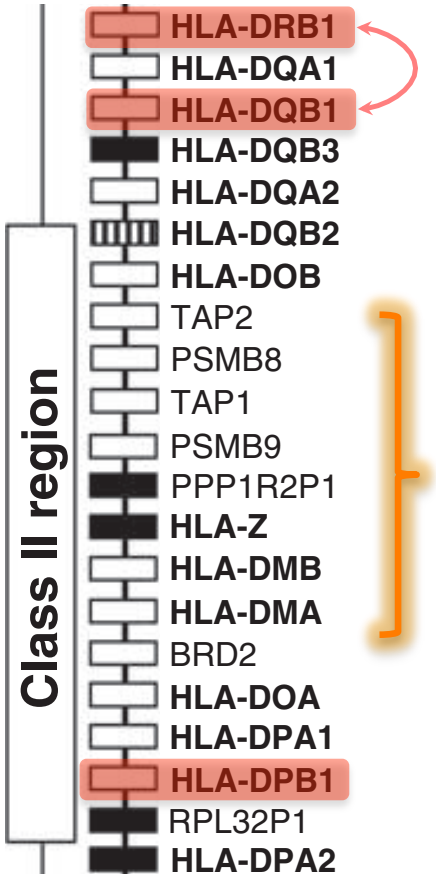



**Figure :** Adapted from Shiina et al, 2009

## Haplotypes in Linkage Disequilibrium

DRB1*13:04 $\sim$ DQB1*03:19

DRB1*11:02:01 $\sim$ DQB1*03:19

DRB1*03:01:01:01 $\sim$ DQB1*02:01:01

–

**Consequence of typing accuracy:**

DPB1*17:01-DQB1*03:01 no more in LD

($DPB1*17:01_{SSO} = 17:01 + 131:01$)

## Global Linkage Disequilibrium between loci

HLA-DRB1 $\sim$ HLA-DQB1

# Hypothesis of gene conversion

**Hill et al (1992)**

*"The DRB1*1304 allele differs by only two amino acids, 57 and 58, from the DRB1*1102 allele, and it was suggested that a small nonreciprocal genetic exchange event, perhaps with a DRB1*0803 allele as the donor, led to the creation of the DRB1*1304 allele."*

```
AA Codon        55                      60
DRB1*11:02:01 CGG CCT GAT GAG GAG TAC
DRB1*13:04    --- --- AGC -CC --- ---
                      Ser Ala
```

# Gene conversion – New insigths

## Donor and recipient

**Recipient:** DRB1*11:02:01

**Donor ?** Not DRB1*08:03

- Presence of specific `Ser-Ala` pattern
- Absent in this study, very rare in Africa

Two potentials donors: DRB1*08:06 & DRB1*13:03:01

- `Ser-Ala` pattern, conserved region, present in Mandenka

Role ?

# Highlights

## Key points

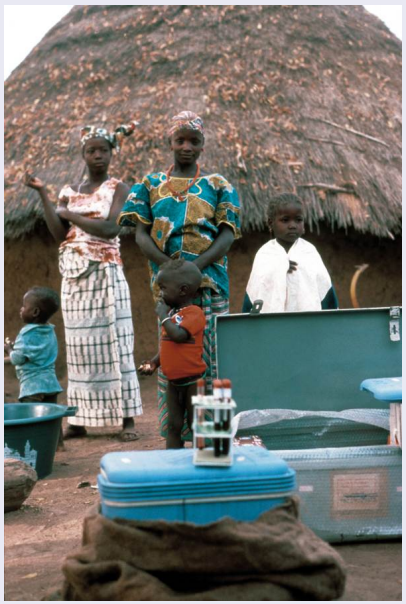
- Importance of accurate typing using NGS and long range sequencing
- High frequency of DRB1*13:04 and DQB1*03:19, **in LD**
- DQB1*03:19 may be involved in *Onchocerciasis* resistance
- Loci HLA-DRB1 and HLA-DQB1 in strong LD
- DRB1*13:04: hypothesis of a gene conversion between DRB1*11:02:01 and either DRB1*08:06 or DRB1*13:03:01

# Acknowledgement

Εφςηαριστό για τιν προσοςή σας

*Thank you for your attention*

## And also many thanks to. . .



The Mandenka people

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